REVERSE ENGINEERING – ACTIVITY REPORT

Product: Portronics Vader Wired Gaming Mouse



Fig. 1: Upper external view of the mouse



Fig. 2: Underside with rating label, PTFE glides and sensor window

PART I – STUDYING THE PRODUCT

1. External Features

Physical Construct

- **Form Factor:** Sculpted, right-handed ergonomic body with a contoured thumb rest.
- **Dimensions (L × W × H):** 120 mm × 80 mm × 40 mm
- **Weight:** ~130 grams
- **Materials & Finish:** Molded white and light-grey ABS plastic with orange accent buttons.
- **Surface Glides:** 4 low-friction PTFE glide pads on the underside.
- **Cable:** 1.5-meter non-detachable USB cable with rubber strain-relief boot.
- **Electrical Ratings:** Model: Vader | Rating: 5V DC, 100mA | Batch: 2604 | Made in China.

Subsystems Used

- Primary Click & Scroll Subsystem (Left/Right Switches, Scroll Wheel)
- Auxiliary Navigation Subsystem (Side Forward/Backward Buttons)
- Sensitivity Control Subsystem (DPI Selector Button)
- Optical Tracking Subsystem (Bottom Optical Sensor & Lens Aperture)
- Ergonomic Shell & Gliding Subsystem (Housing, Thumb Rest, PTFE Feet)
- Power & Wired Data Interface Subsystem (USB Wiring Harness)

Description of Subsystems

Subsystem	Description
Primary Click & Scroll	Micro-switch activated buttons for standard OS clicks and a rubberized wheel for vertical scrolling and middle-click.

Subsystem	Description
Auxiliary Navigation	Two side-mounted orange switches for browser forward/backward actions or in-game hotkeys.
Sensitivity Control	Central orange DPI switch behind the scroll wheel for on-the-fly resolution toggling.
Optical Tracking	Underbody optical sensor window and LED illuminating the surface to calculate motion.
Ergonomic Shell & Gliding	ABS plastic outer chassis protecting internal components and ensuring smooth desktop motion.
Power & Data Interface	Direct USB connection delivering 5V DC power (100mA) and zero-latency input signaling.

Modes of Operation

- **Standard Navigation Mode:** Default OS/browser productivity mode.
- **On-the-Fly DPI / Precision Mode:** Instant toggling between high-precision (editing/aiming) and high-speed cursor tracking.
- **Gaming / Custom Macro Mode:** Utilizing thumb side keys and middle-click inputs mapped to custom software commands.

2. Purpose of the Product

- Serves as a low-latency Human Interface Device (HID) converting mechanical hand movements and button inputs into digital cursor commands on a computer.

3. User Needs Addressed

User Need	How the Product Addresses It
Ergonomics & Comfort	Thumb rest and contoured shape reduce palm fatigue during extended sessions.
Immediate Sensitivity Adjustment	Hardware DPI toggle eliminates needing OS-level software adjustments.
Reliability & Low Latency	Wired USB connection removes input lag, battery depletion, and wireless interference.
Enhanced Productivity	Side thumb switches accelerate file and web navigation.

PART II – PRODUCT ORIGIN TO LATEST INVENTIONS

1. Difference in Technology Across Versions

Generation	Improvements	Drawbacks
1.1 Early Stage (Traditional Mechanical Ball Mice)	Basic 2D cursor tracking using a rubber-coated steel roller ball and internal encoder wheels.	Heavy dust accumulation on rollers causing skipping; low polling rate and DPI (100–400 DPI).
1.2 First Generation (Basic Red Optical Mice)	Solid-state optical CMOS image sensors with red LED illumination; no moving mechanical tracking parts.	Poor tracking on glass/reflective surfaces; fixed resolution (400–800 DPI); no hardware DPI toggling.
1.3 Second Generation (High-DPI Gaming & Multi-Button Optical Mice)	Hardware DPI switches (800–3200+ DPI), multi-button layouts, right-handed contours, and integrated microcontrollers.	Basic monochrome lighting; higher USB power draw; potential tracking jitter at high accelerations.
1.4 Latest Generation (Modern Smart Gaming Mice – e.g., Portronics Vader)	Single-chip SoC architecture (Sunplus SPCP190), multi-zone RGB lighting systems, high polling rate (up to 1000Hz), thumb skirts, and PTFE glides.	Higher structural design complexity; sensitivity to ESD without proper PCB ground planes.

2. Current Status – Core Smart Technologies Used

- **Single-Chip Optical SoC Architecture:** Integrates CMOS image sensor, DSP engine, debouncing logic, and USB MCU into a compact package.
- **Dynamic Optical Tracking Engine:** Captures surface image frames at high rates (7000+ FPS) to compute sub-millimeter motion deltas.
- **Hardware DPI Quantization:** Instant resolution adjustment via microswitch without driver software dependency.
- **Multi-Zone RGB Light Mapping:** Secondary lighting controller driving surface-mount LEDs (SMD LED1–6) via light diffusers.

3. Engineering Importance of This Evolution

- **Human Ergonomics:** Sculpted palm support reduces repetitive strain injuries (RSI) during extended use.
- **Manufacturing Cost Optimization:** Single-chip SoC reduces overall BOM count and simplifies PCB manufacturing.
- **Performance & Reliability:** Solid-state optoelectronics extend device lifespan and eliminate input lag.

PART III – INTERNAL COMPONENTS & WORKING FLOW

1. Introduction of the Product

- The Portronics Vader internal design centers around a compact single-sided printed circuit board (PCB) powered by an integrated optical System-on-Chip (SoC).

2. Photograph of the Internal Components of the Product

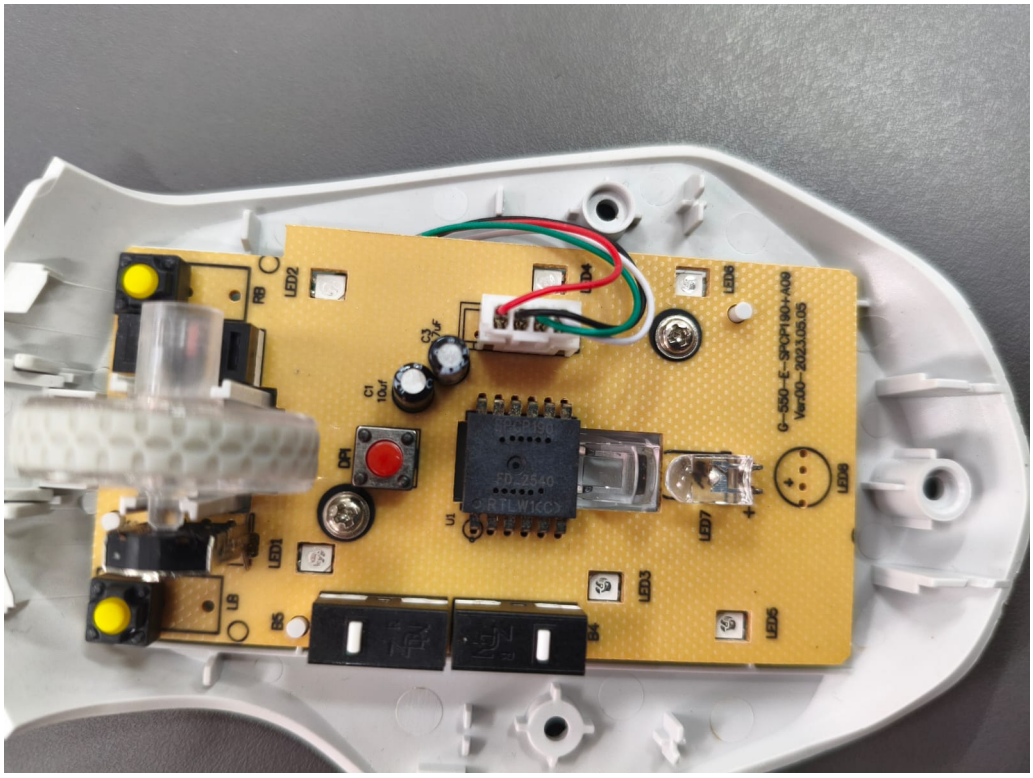


Fig. 3: Main PCB – U1 (SPCP190), LED7, switches, capacitors C1/C3 and USB harness

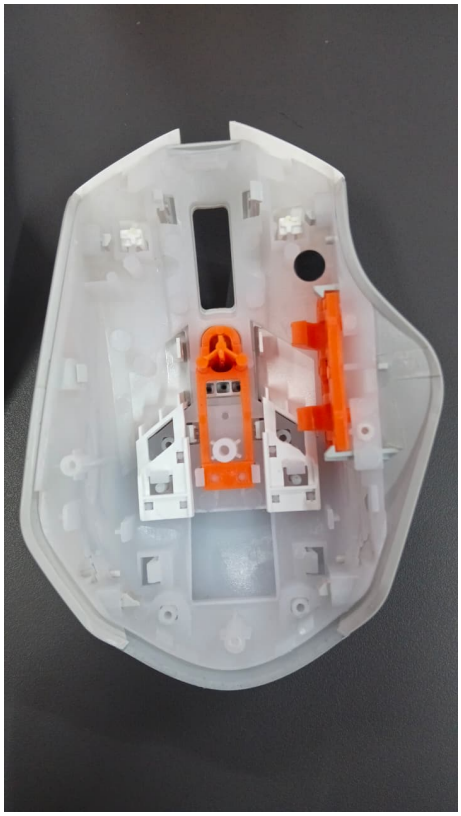


Fig. 4: Upper shell – orange button actuators and mounting posts

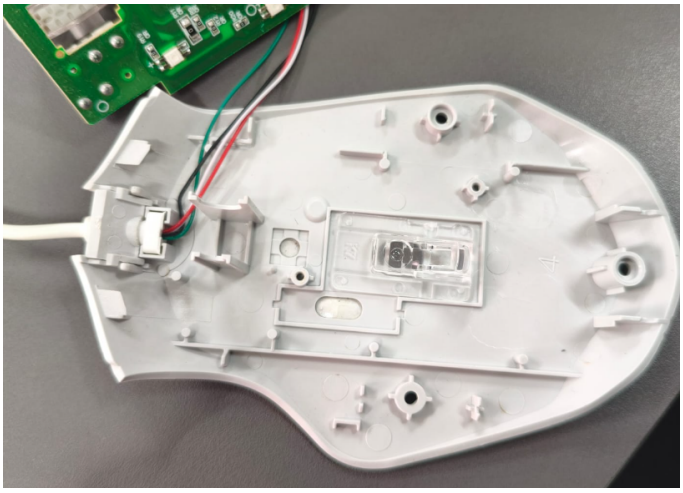


Fig. 5: Lower shell – transparent lens window for the optical sensor, cable entry and mounting posts

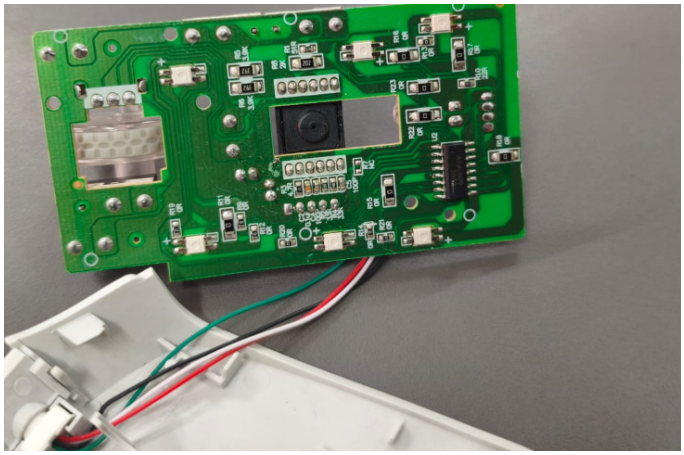


Fig. 6: PCB underside – sensor window, U2 IC, SMD LEDs and resistors

3. Internal Block Diagram (Functional View)

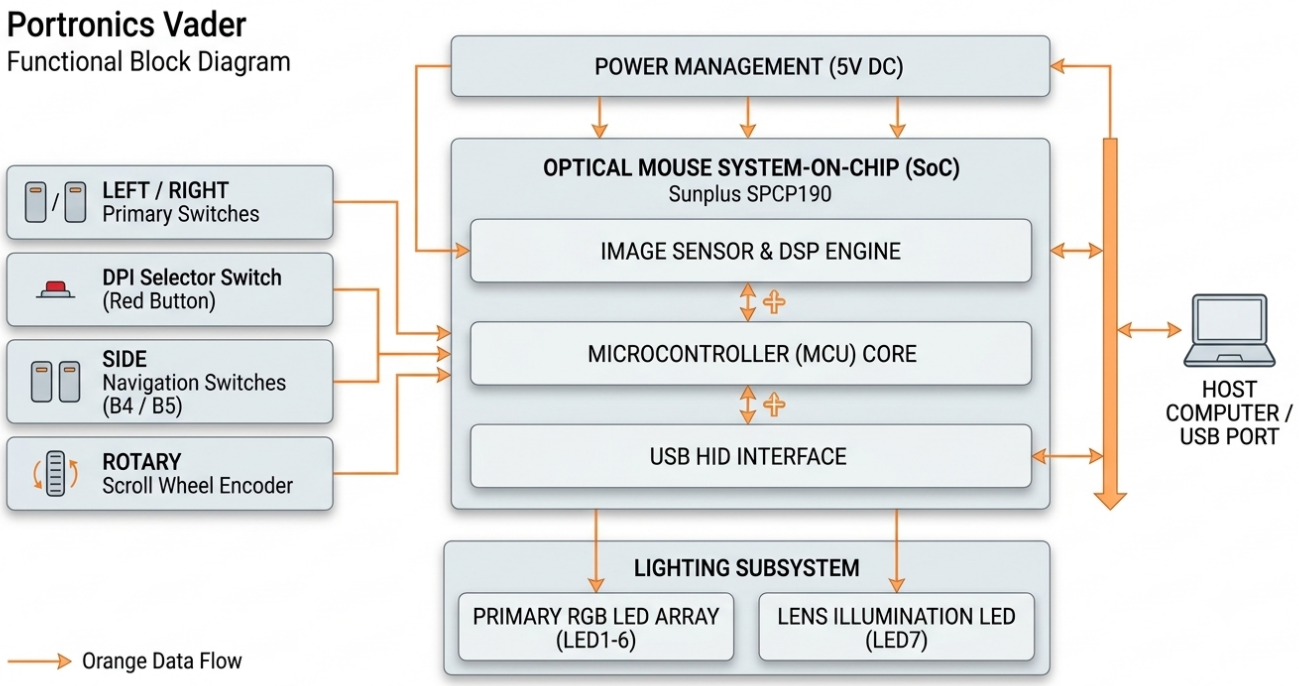


Fig. 7: Functional block diagram of the Portronics Vader

4. Internal Subsystems with Detailed Explanation

4.1 Primary Optical Tracking Subsystem

Aspect	Details
Function	Captures high-speed surface reflectivity data to compute 2D motion coordinates.
Working Principle	Red LED illuminates the surface; light bounces through an acrylic prism lens into the CMOS optical array inside the Sunplus SPCP190 SoC.
Characteristics	Zero moving tracking parts, frame rate up to 7000+ FPS, highly responsive across opaque mouse-pad and desk surfaces.

4.2 Click and Scroll Input Subsystem

Aspect	Details
Function	Detects left, right and middle clicks and vertical scrolling.
Working Principle	Left/right micro-switches close on press and are read by the SoC. The scroll wheel turns a rotary encoder whose pulses indicate direction and distance.
Characteristics	Mechanical micro-switches with tactile feedback; rubberized scroll wheel; encoder output read directly by the SoC.

4.3 Auxiliary and DPI Control Subsystem

Aspect	Details
Function	Provides forward/backward navigation (B4/B5) and on-the-fly DPI selection.
Working Principle	Side switches and the tactile DPI switch connect to SoC input pins; a press cycles the stored resolution setting inside the chip with no host driver needed.
Characteristics	Hardware-level DPI cycling; two programmable thumb buttons; the DPI state is also reflected through lighting feedback.

4.4 Lighting Subsystem

Aspect	Details
Function	Provides surface illumination for tracking and RGB aesthetic lighting.
Working Principle	LED7 lights the tracking surface. SMD LEDs LED1–6 are driven by the SoC lighting outputs and shine through the shell diffusers.
Characteristics	Primary RGB LED array (LED1–6) plus a dedicated lens illumination LED (LED7); low current draw within the 100mA rating.

4.5 Power and USB HID Interface Subsystem

Aspect	Details
Function	Supplies 5V DC power and carries data between the mouse and host computer.
Working Principle	The 4-core USB cable (red = VCC, black = GND, green/white = data pair) feeds the PCB through a connector. Decoupling capacitors (C1, C3) stabilize the supply; the SoC's USB HID interface reports movement and button data.
Characteristics	Plug-and-play HID class device, 5V / 100mA, up to 1000Hz polling, no battery and no wireless interference.

5. Component Function Table

Table 1: Component Function Table – Portronics Vader (G-550-E main PCB)

Component Ref.	Component Name	Subsystem	Specific Function
U1 (SoC)	Sunplus SPCP190 optical mouse SoC	Optical Tracking & USB MCU	Integrates the 7000+ FPS image sensor, DSP engine, debouncing logic and full USB HID interface.
LB / RB Microswitches	Left / right micro-switches	Primary Click	Register left and right clicks; contact bounce is filtered by the SoC.
DPI Switch (red)	Tactile DPI selector switch	Sensitivity Control	Cycles the tracking resolution on the fly without driver software.
B4 / B5 Side Switches	Side navigation micro-switches	Auxiliary Navigation	Provide browser forward/backward or in-game hotkey inputs.
Scroll Encoder	Rotary scroll wheel encoder	Scroll Sensing	Converts wheel rotation into direction and step pulses; wheel press gives middle-click.
LED7 (Tracking)	Surface illumination LED	Optical Tracking	Illuminates the desk surface so the CMOS sensor can capture image frames.
LED1–6 (Perimeter)	SMD RGB LEDs	Lighting	Provide multi-zone RGB lighting through the shell light diffusers.
U2 (Lighting IC)	16-pin IC on PCB underside	Lighting	Assisting controller for the lighting circuit (function inferred from board position).
C1 / C3	Electrolytic capacitors	Power	Smooth and decouple the 5V supply to the SoC and LEDs.
Strain Relief Boot	Rubber cable boot	Cable / Housing	Protects the USB cable at the point where it enters the shell.

6. Working Flow (Summary)

- The host USB port supplies 5V DC, which powers the SoC, sensor illumination and lighting.
- The optical sensor and DSP engine compare successive surface images to compute X/Y motion.
- Button, DPI and scroll inputs are scanned by the MCU core, which packages motion and button data into HID reports.
- The USB HID interface sends the reports to the computer, which moves the cursor or triggers the mapped action.

PART IV – FUNCTIONAL PROTOTYPE & SIMULATION

1. Objective of the Prototype

After studying the internal working of the Portronics Vader (Parts I–III), a functional model of its main input features was built and simulated on **Tinkercad Circuits** using an **Arduino Uno**. The model reproduces, in simplified form, the behaviour of the mouse subsystems: left/right clicks, hardware DPI cycling with visual feedback, a scroll input and a motion sensor whose output is scaled by the selected DPI. The aim is to verify the working principle of the product before relating it to the real single-chip SoC.

2. Mapping: Real Mouse vs. Simulation Model

Each subsystem of the real mouse (Table 1, Part III) is represented by a readily available prototyping component:

Vader Subsystem	Real Component	Simulation Component	Arduino Pin
Primary Click (Left)	Left micro-switch (LB)	Push button	D2 (INPUT_PULLUP)
Primary Click (Right)	Right micro-switch (RB)	Push button	D3 (INPUT_PULLUP)
Sensitivity Control	Red tactile DPI switch	Push button	D4 (INPUT_PULLUP)
DPI / Lighting Feedback	SMD LEDs (LED1–6)	Red LED + series resistor	D9 (PWM output)
Optical Tracking	SPCP190 CMOS sensor + LED7	PING))) ultrasonic sensor	D7 (SIG, 3-pin)
Scroll Sensing	Rotary scroll encoder	Potentiometer	A0 (analog input)
Power & USB Interface	5V USB harness, SoC MCU	Arduino Uno (USB, 5V)	5V / GND / USB

3. Circuit Diagram (Tinkercad)

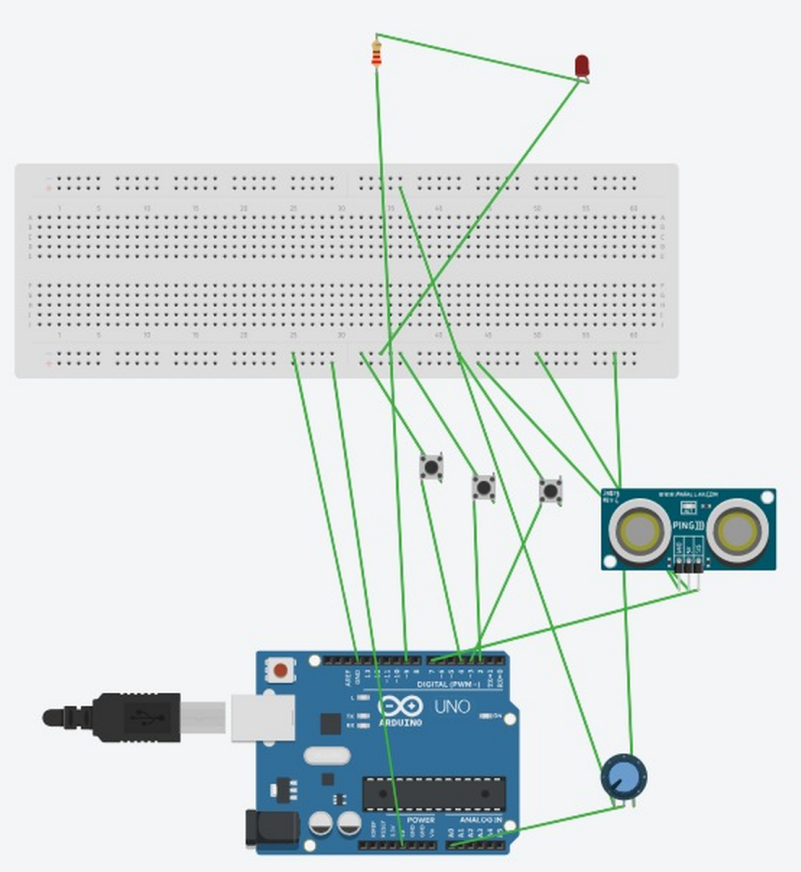


Fig. 8: Tinkercad circuit – Arduino Uno with three push buttons, PING))) sensor, potentiometer, LED with resistor and breadboard power rail

4. Components Used

Component	Qty	Role in the Model
Arduino Uno R3	1	Acts as the SoC / MCU core: reads inputs, processes data, sends output over USB serial.
Push buttons	3	Left click, right click and DPI selector inputs (active LOW using internal pull-ups).
PING))) ultrasonic sensor	1	Stand-in for the optical tracking sensor; returns a distance value used as movement.
Potentiometer	1	Stand-in for the scroll-wheel encoder; gives a 0–1023 analog scroll value.
Red LED + resistor	1	DPI level indicator; brightness rises with the selected DPI.
Breadboard & jumper wires	–	Common 5V / GND distribution and connections.

5. Working of the Program

The program runs in a continuous loop and mirrors the working flow described in Part III (Section 6):

Step	Operation	Equivalent in Real Mouse
1. DPI Handling	A falling edge on D4 cycles dpiState through 800, 1600 and 3200 DPI. A 200 ms delay acts as software debounce. The new value is printed on the serial monitor.	Red DPI switch read by the SoC; resolution cycles without driver software.
2. LED Feedback	analogWrite(D9, dpiState x 80) sets the LED to three brightness levels (80, 160, 240).	DPI state reflected through the lighting subsystem.

Step	Operation	Equivalent in Real Mouse
3. Motion Sensing	A 5 microsecond trigger pulse is sent to the PING))) sensor and the echo time is converted to distance: cm = duration x 0.034 / 2. Position delta = distance x dpiState.	CMOS sensor and DSP compute X/Y motion; higher DPI gives larger counts for the same movement.
4. Scroll Reading	analogRead(A0) returns 0–1023 from the potentiometer.	Rotary encoder pulses give scroll direction and distance.
5. Click Detection	D2 or D3 reading LOW prints LEFT CLICK! or RIGHT CLICK!.	Micro-switches close on press and are read by the SoC.
6. Output Report	Position delta and scroll value are printed every 200 ms at 9600 baud.	USB HID report sent to the host computer (up to 1000 Hz).

DPI Levels Used in the Model

dpiState	DPI Value	LED PWM (D9)	Position Delta (for distance d)
1	800 DPI	80	1 x d
2	1600 DPI	160	2 x d
3	3200 DPI	240	3 x d

6. Arduino Source Code

The complete program uploaded to the Arduino Uno in Tinkercad is given below.

```
const int TRIG_PING = 7;
const int BTN_LEFT = 2;
const int BTN_RIGHT = 3;
const int BTN_DPI = 4;
const int POT_SCROLL = A0;
const int LED_DPI = 9;

int dpiState = 1; // 1: 800 DPI, 2: 1600 DPI, 3: 3200 DPI
bool lastDpiBtnState = HIGH;

void setup() {
  Serial.begin(9600);

  pinMode(BTN_LEFT, INPUT_PULLUP);
  pinMode(BTN_RIGHT, INPUT_PULLUP);
  pinMode(BTN_DPI, INPUT_PULLUP);

  pinMode(LED_DPI, OUTPUT);
}

void loop() {
  // 1. DPI Switch Handling
  bool currentDpiBtn = digitalRead(BTN_DPI);
  if (lastDpiBtnState == HIGH && currentDpiBtn == LOW) {
    dpiState = (dpiState % 3) + 1;
    Serial.print("DPI Changed to: ");
    Serial.println(dpiState * 800);
    delay(200);
  }
  lastDpiBtnState = currentDpiBtn;

  analogWrite(LED_DPI, dpiState * 80);

  // 2. Sensor Reading (PING))) 3-pin)
  pinMode(TRIG_PING, OUTPUT);
  digitalWrite(TRIG_PING, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PING, HIGH);
  delayMicroseconds(5);
  digitalWrite(TRIG_PING, LOW);

  pinMode(TRIG_PING, INPUT);
  long duration = pulseIn(TRIG_PING, HIGH);
  int distanceCm = duration * 0.034 / 2;
  int scaledMovement = distanceCm * dpiState;

  // 3. Scroll Wheel Reading
  int scrollVal = analogRead(POT_SCROLL);

  // 4. Output Logs
  if (digitalRead(BTN_LEFT) == LOW) {
    Serial.println("LEFT CLICK!");
  }
  if (digitalRead(BTN_RIGHT) == LOW) {
    Serial.println("RIGHT CLICK!");
  }

  Serial.print("Position Delta: ");
  Serial.print(scaledMovement);
  Serial.print(" | Scroll Value: ");
  Serial.println(scrollVal);

  delay(200);
}
```

Listing 1: Arduino sketch for the Vader functional model (DPI, clicks, motion and scroll)

7. Serial Monitor Output Format

On every cycle the program prints one status line, and prints extra lines when an event occurs. The messages produced by the code have the following format:

Event	Serial Monitor Message
DPI button pressed	DPI Changed to: 1600
Left button held	LEFT CLICK!
Right button held	RIGHT CLICK!
Every 200 ms cycle	Position Delta: *value* | Scroll Value: *0–1023*

8. Observations & Limitations of the Model

- **Sensor analogy:** the real mouse uses a CMOS image sensor to find X/Y motion on a surface. The PING))) sensor measures distance in one direction only, so it represents tracking in a simplified way.
- **Scroll analogy:** the real wheel uses a rotary encoder producing pulses; the potentiometer gives a continuous analog value instead.
- **Debounce:** the real SoC filters switch bounce in hardware; the model uses a 200 ms software delay, which also limits the loop rate to about 5 readings per second (the real mouse polls at up to 1000 Hz).
- **Output:** the real mouse sends USB HID reports; the model prints text to the serial monitor.
- **Possible improvements:** add a timeout to pulseIn() so the loop never blocks if no echo is received, and use millis()-based debouncing so the loop can run faster.

9. Conclusion

The reverse engineering study showed that the Portronics Vader is built around a single-chip optical SoC (Sunplus SPCP190) that combines the image sensor, DSP engine, button scanning and USB HID interface, supported by micro-switches, a scroll encoder, a DPI switch and an LED lighting subsystem. The Tinkercad prototype demonstrates the same working flow – inputs are read, DPI scales the motion data, lighting gives feedback and the results are reported to the host – using an Arduino Uno. Building the model confirmed how the subsystems cooperate and showed why integrating them in one chip lowers cost, size and latency in modern gaming mice.